

Department of Computer Science and Engineering
SAGE University
Indore 452020, Madhya Pradesh, India

The Editors
Journal of Seismology
Springer Nature

Dear Editors,

We submit for your consideration our manuscript, “A versioned, leakage-audited USGS *Did You Feel It?* benchmark for machine-learning evaluation of severe felt intensity.” We request handling as a data or software paper documenting a public seismological resource; if that label is unavailable in Snapp, we request consideration as an Original Paper.

Machine-learning studies of macroseismic felt intensity commonly use bespoke datasets and random record-level splits. Their results can be difficult to reproduce and vulnerable to leakage between related earthquake events. Our manuscript addresses this problem by releasing a frozen, single-source USGS DYFI benchmark, a six-category leakage audit, outcome-independent sequence-grouped roles, fixed baselines, calibration-aware scoring, and a protected temporal evaluation.

The frozen $M \geq 5$ snapshot contains 7,520 source records, 2,362 eligible events, and a 430-event temporal holdout. The observed random-versus-sequence split differences are small and reported descriptively; no winning model, operational forecasting benefit, causal effect, or cross-system generalization is claimed. All deviations and unexecuted branches are disclosed.

The work fits the *Journal of Seismology* as an observational macroseismology and citizen-science data/software resource. It directly aligns with the journal’s stated interest in machine learning, seismological datasets, software tools, and publicly available resources supporting transparent, reproducible earthquake science.

Data, code, frozen results, and reproduction instructions are available at <https://github.com/aksha74/dyfi-benchmark-reliability>. A clean clone reproduces the 15-page manuscript, passes all 104 manuscript checks and 110 benchmark tests, and reconstructs the recorded results and holdout predictions.

The manuscript is original, is not under consideration elsewhere, and has not been published previously. The authors declare no competing interests. The study uses aggregate public-domain event metadata, involves no human participants, and redistributes no respondent-level identifying information. Generative-AI assistance is disclosed in Methods and all assisted work was verified by the authors.

Sincerely,

Akshay Sharma (corresponding author) and Lalji Prasad
Department of Computer Science and Engineering
SAGE University, Indore, India
aksha74@gmail.com